

2022-Ph-H1-Q24

Section: Electricity

Topic: Capacitors

Which of the following increases the capacitance of a capacitor?

Answer: A

(Increasing the area of the plates)

Guidance for Students:

Capacitance depends on plate area, separation, and dielectric:

$$C = \epsilon \frac{A}{d}$$

Revision Tips:

- Increase plate **area** $\rightarrow$ increases C
- Increase **distance** $\rightarrow$ decreases C
- Dielectric material increases $\epsilon \rightarrow$ increases C